DELHI TECHNOLOGICAL UNIVERSITY
(Formerly Delhi College of Engineering)
Shahbad Daulatpur Village, Rohini
Delhi, India

UGV-DTU

Avni



Submission Date , July 2, 2026

Competition	UGVC-10 , 10th International Competition, MTC Cairo, Egypt
Faculty Advisor	Prof. S. Indu, Dean (Digital Education), DTU
Team Captain	Vedant Singh , DTU

I hereby certify that the design and development of the vehicle **Avni**, described in this report, represents a substantial and significant engineering effort by the student team in preparation for this competition.

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1 Team Organization and Design Process

1.1 Introduction

UGV-DTU is a multidisciplinary undergraduate student team from Delhi Technological University (DTU), combining expertise in software engineering, mechatronics, embedded systems, and computer vision to design and validate autonomous ground vehicles for international competition. **Avni** is the team's most ambitious system to date, a differential-drive rover on a modular aluminium chassis, powered by a dual battery system consisting of one 6S LiPo and one 6S $LiFePO_4$ batteries, sensing its environment through a 2D LiDAR, dual cameras, IMU, and GPS, and driven by a ROS2 Jazzy autonomy stack with integrated perception, sensor fusion, path planning, and autonomous face recognition via a pan-tilt turret. Beyond competition, Avni serves as a hands-on research platform with methodologies designed for transfer to junior members, ensuring long-term technical continuity.

1.2 Organisation

The team is structured across four functional groups i.e. Software, Mechanical Electronics and Corporate, with clearly delineated responsibilities and a designated Team Captain accountable for all competition-day decisions.

Name	Dept.	Year	Hours	Name	Dept.	Year	Hours
Vedant Singh*	ME	3rd	750+	Lakshya Jindal	IT	2nd	400+
Yuvraj Gupta	EE	3rd	750+	Ankit Roy	EE	1st	500+
Akshat Rajwansh	ME	3rd	750+	Prakhar Mishra	ME	1st	500+
Gatik Gupta	MAM	3rd	600+	Nahar Singh	ME	1st	500+
Narayan Mishra	ENE	2nd	700+	Sanchit Kumar	EP	1st	400+
Daksh Panchal	CSE	2nd	700+	Ayush Kumar	ECE	1st	300+
Manaan Tyagi	EE	2nd	700+	Ananya Tripathi	BT	2nd	200+
Harshit Tyagi	EE	2nd	400+	Apeksha Singh	BT	2nd	200+
Dhruv Bansal	CSE	2nd	400+	Bhuvi Malhotra	BT	2nd	200+
Kartikay Tayal	IT	2nd	100+	Anusri Yadav	ENE	1st	100+

Table 2: Member Contribution Sheet

1.3 Design Assumptions and Design Process

Development was governed by a star-model systems engineering approach, in which each phase was paired with a dedicated verification step before integration proceeded. Phase 1 trade studies evaluated competing options across vehicle configuration, sensor suite, compute platform, power architecture, and navigation strategy, while simultaneously defining subsystem interfaces, communication protocols, coordinate frames, and power budgets. Software development followed a branch-per-subsystem Git workflow, ensuring code-level stability throughout parallel development. Hardware assembly mandated complete power isolation between the electronics and drivetrain batteries, with independent emergency-stop mechanisms verified before any powered testing commenced. Phase 3 simulation in Gazebo Harmonic served as a validation gate prior to hardware deployment, allowing navigation and perception algorithms to be stress-tested without mechanical risk. Final verification encompassed not only nominal mission tasks but also deliberate fault-injection testing, confirming that the system degrades safely under sensor failures, software faults, and communication interruptions.

1.4 Budget and Cost Accounting

Avni is funded through Delhi Technological University's institutional research allocation. SolidWorks and Ansys software licences are provided through educational sponsorship.

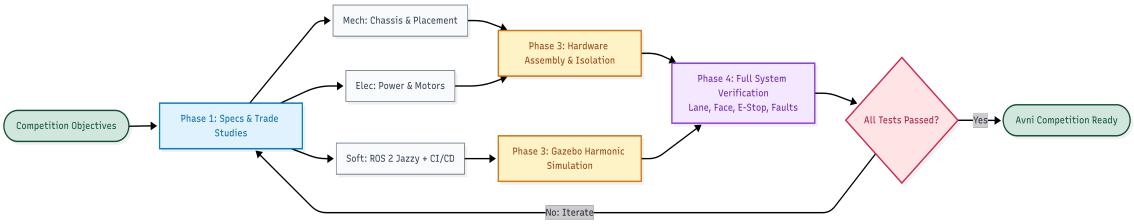


Figure 1: Star-Model Engineering Design Process

Item	Specification	Quantity	Total Cost
Chassis (3030 Al T-Slot)	6061 Al Extrusion	1	,
Onboard Computer	[Model / Processor / RAM]	1	,
360° LiDAR	RPLiDAR A3M1 , 10 Hz, 25 m	1	,
Front Camera	640×480 @ 30 Hz, 60° HFoV	1	,
Turret Camera	640×480 @ 30 Hz, 60° HFoV	1	,
IMU	6-DoF, 100 Hz, I²C/USB	1	,
GPS Module	GenX RTK F9P Tough Ground RTK GPS, 5 Hz NMEA	1	,
Motor Controllers	Encoder feedback	2	,
Drive Motors	DC geared, high-torque	2	,
Pan-Tilt Servos	Digital ±170°/±80°	2	,
Laser Pointer	<1 mW class	1	,
Arduino Uno (turret ctrl)	USB serial, 115200 baud	1	,
Battery Pack	24 V, [X] Ah Li-ion	1	,
Outer Body (Acrylic/Al)	4 mm acrylic sheet	,	,
Wheels & Hardware	,	,	,
Miscellaneous (cables)	,	,	,
TOTAL			

1.5 Project Timeline

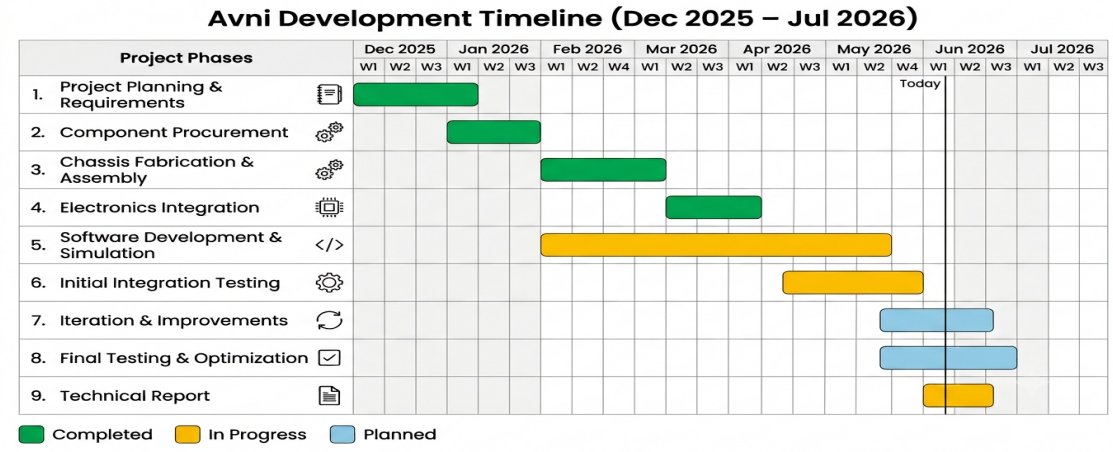


Figure 2: Gantt Chart – Avni Development Timeline (Dec 2025 – Jul 2026)

1.6 Safety Measures

Avni incorporates multiple hardware-based safety mechanisms to ensure reliable operation in both autonomous and manual modes. A dual emergency stop system consists of a normally-closed mechanical E-stop switch that disconnects motor power through a hardware relay in under 10 ms, and an independent LoRa-based wireless E-stop capable of remotely disabling the vehicle from distances exceeding 150 m. Both mechanisms operate entirely at the hardware level, independent of the ROS 2 software stack, ensuring fail-safe operation even in the event of software failure. Additionally, a NeoPixel LED status indicator provides clear visual feedback on the vehicle's operating state, displaying a solid illumination during startup and standby, blinking continuously during autonomous navigation to alert nearby personnel, and returning to a solid state once the mission is complete.

2 Technical Innovations

Avni incorporates several purpose-built engineering innovations spanning mechanical design, electronics architecture, and autonomous software that collectively distinguish it from conventional rover platforms. On the mechanical side, a passive rocker differential suspension mechanism maintains continuous four-wheel ground contact across uneven terrain without active actuators or electronic intervention, delivering inherent traction and stability. This is complemented by custom additively manufactured TPU 95A wheels featuring a 30% gyroid lattice infill, which provide puncture immunity and terrain-induced vibration absorption while eliminating the need for a conventional spring–damper system, resulting in a lightweight, compliant drivetrain that requires no mechanical suspension maintenance.

The electronics architecture is centered around a custom Power Distribution and Logic Control Board that integrates power regulation, embedded control, signal conditioning, protection circuitry, and subsystem interfacing into a single compact platform, significantly reducing wiring complexity and improving system reliability. An EMI-aware mixed-signal PCB layout isolates high-current drivetrain circuits from sensitive digital electronics using dedicated ground planes, wide copper pours, decoupling networks, and optimized trace routing. Reliability is further enhanced through electrically isolated dual-battery domains, with a 6S LiPo powering onboard electronics and a 6S $LiFePO_4$ battery dedicated to propulsion, while deterministic real-time control tasks are offloaded to a Teensy 4.1 microcontroller, allowing the onboard computer to focus exclusively on perception and autonomous navigation.

At the software level, Avni employs a dynamic lane-constrained local goal generation strategy for autonomous navigation. Rather than driving directly to the final waypoint, the system continuously extracts the forward drivable region from a bird's-eye-view image, identifies the lane centreline, and projects candidate navigation points into world coordinates. These candidates are evaluated using occupancy costmaps and LiDAR to ensure adequate obstacle and pothole clearance before selecting the safest intermediate sub-goal approximately 2-4 m ahead. This receding-horizon planning process repeats continuously as the vehicle advances, maintaining lane centering while adapting in real time to changing road geometry and obstacles, resulting in smoother and more reliable autonomous navigation than conventional waypoint-based planning.

3 Description of Mechanical Design

3.1 Design Philosophy

Avni's mechanical design is governed by three principles: **modularity** – each sensor, compute module, and electronic component can be removed and replaced without disturbing adjacent assemblies; low centre of gravity – heavy components (battery pack, motor controllers) are located at the lowest chassis tier for dynamic stability during cornering; and **full accessibility**– all end connectors, fuses, and maintenance interfaces are reachable from the top surface without body panel removal.

3.2 Chassis Design

The rover chassis is fabricated from **Aluminium Alloy 6061** sheet, selected for its high strength-to-weight ratio, corrosion resistance, and manufacturability. Structural members are laser-cut with high dimensional accuracy and subsequently CNC bent to achieve the required geometry while minimizing part count. The assembly employs M6 bolted joints, enabling rapid assembly, field-serviceability, and straightforward component replacement without permanent structural modifications.

The internal architecture is divided into two isolated compartments by the centrally mounted suspension cross-rod, which also contributes to structural rigidity, trategically positioned ventilation cut-outs facilitate forced-air cooling of the onboard computer and power electronics, improving thermal dissipation during sustained operation. The modular chassis architecture provides high accessibility to subsystems, simplifies maintenance, reduces downtime, and allows future hardware upgrades with minimal redesign.

3.3 Vehicle Dimensions and Configuration

Avni is a four-wheeled differential-drive platform. Key parameters:

- Chassis body: 1.3 m × 0.85 m × 1.2 m , within the competition's permitted bounds (90–200 cm length, 60–120 cm width, ≤120 cm height).
- Wheel radius: 240mm

Component	Mass (kg)	Mass Fraction (%)
Aluminium Chassis Frame	8	13%
Drive System	33	54%
Wheels and Hub	6	10%
Battery Pack	7	11.5%
Onboard Computer & Electronics	7	11.5%
Total	**61**	**100**

3.4 Differential Mechanism

Avni employs a **passive differential bar (rocker-bogie differential)** as its chassis articulation mechanism, inspired by **NASA's Mars Exploration Rovers**. This mechanism enables all four wheels to maintain continuous ground contact on uneven terrain without requiring active suspension. The differential bar consists of a rigid **SS 304 beam** pivoted at the chassis center and linked to the left and right rocker assemblies. As one side moves over an obstacle, the bar transfers a proportional counter-rotation to the opposite side, reducing chassis tilt and improving sensor stability, obstacle traversal, and overall reliability through a completely passive design.

3.5 Custom Wheel Design

The rover employs custom-designed wheels additively manufactured from **Thermoplastic Polyurethane (TPU 95A Shore hardness)** at 30% gyroid infill density. This represents a significant departure from conventional pneumatic or solid rubber wheels. The key innovations are:

1. **Puncture Immunity:** Being a solid elastomeric structure with no pressurised cavity, the wheel cannot sustain puncture failure – a critical advantage in rocky and uneven terrains competition terrain.
2. **Integrated Compliant Suspension:** The inherent viscoelastic properties of TPU 95A, combined with the controlled void geometry of the 30% infill lattice, enable the wheel body itself to deform elastically under load, absorbing micro-terrain vibrations. This passive compliance reduces shock loads transmitted to the drivetrain and chassis, partially substituting for a conventional spring-damper suspension stage.

Wheels are fabricated via Fused Deposition Modelling (FDM) using a 0.4 mm nozzle at 0.2 mm layer height, with a gyroid infill pattern selected for its isotropic mechanical response.

3.6 Structural Analysis / FEA

Finite Element Analysis was performed in ANSYS to validate the structural integrity of the two primary load-bearing components: the chassis frame and the custom TPU lattice wheel. Both were evaluated under worst-case static loading conditions representative of full payload mass combined with dynamic cornering and obstacle-impact loads, with results assessed against a minimum target safety factor of 2.5, consistent with the joint-level safety factor established analytically for the wheel hub.

3.6.1 Chassis Structural Analysis

The chassis frame was modelled with fixed constraints at the rocker pivot mounts and loaded with the full vehicle payload distributed across the two compartments (Section 3.2), with an additional lateral load case applied to simulate cornering inertia. The analysis evaluates von Mises stress distribution, total deformation, and factor of safety across the aluminium 6061 structure, with particular attention to the differential bar pivot housing and turret mount cross-member, identified as the highest-stress regions due to concentrated reaction loads.

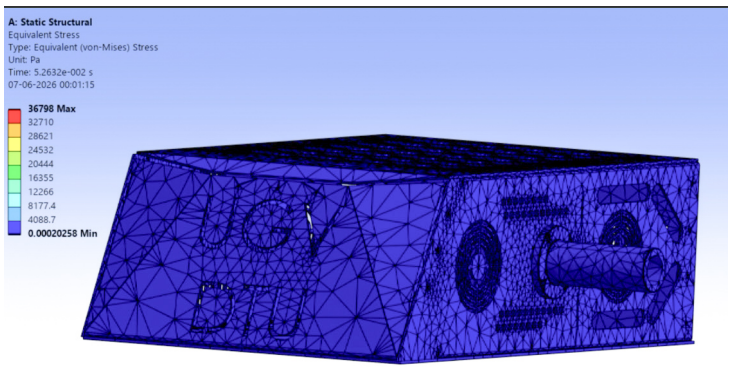


Figure 3: FEA Results – Chassis Frame under Full Payload Load

3.6.2 TPU Wheel Structural Analysis

The custom TPU 95A wheel, including the 30% gyroid infill lattice and steel-insert hub interface introduced in Section 3.5, was analysed under peak motor stall torque and full radial payload load to verify the safety factor independently of the analytical hand calculation. The hyperelastic material behaviour of TPU 95A was approximated in ANSYS using an appropriate elastomeric material model to capture the wheel's compliant deformation response under load.

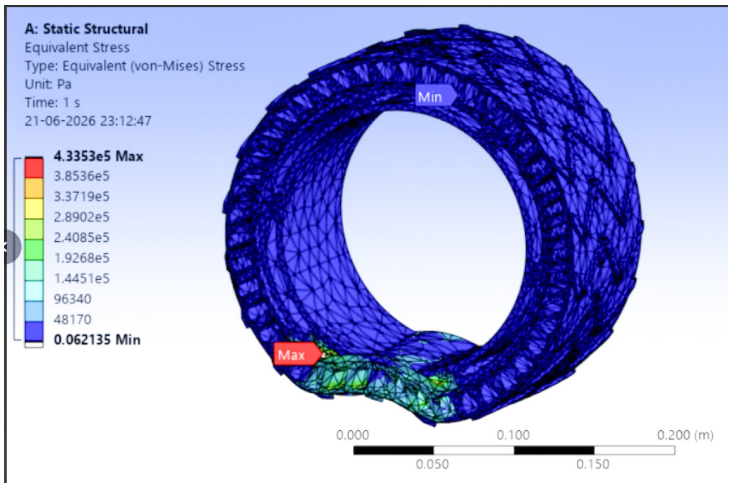


Figure 4: FEA Results – TPU Wheel under Full Payload Load

4 Description of Electronic and Power Design

4.1 Overview

The rover employs a centralized custom-designed **Integrated Power Distribution and Logic Control Board** that integrates power regulation, embedded control, signal conditioning, system protection, and electrical distribution into a compact platform, reducing wiring complexity while improving reliability and maintainability. A **distributed processing architecture** separates real-time control from

high-level computation. A **Teensy 4.1** microcontroller handles motor control, encoder processing, PID computation, and actuator interfacing, while the onboard computer executes **ROS 2** navigation, perception, and mission planning. The electrical system uses two isolated power sources: a **6S LiPo** battery for onboard electronics and a **6S LiFePO$_4$** battery for the high-current drivetrain. Fused power distribution, battery monitoring, and protection circuitry improve system safety, reliability, and fault tolerance.

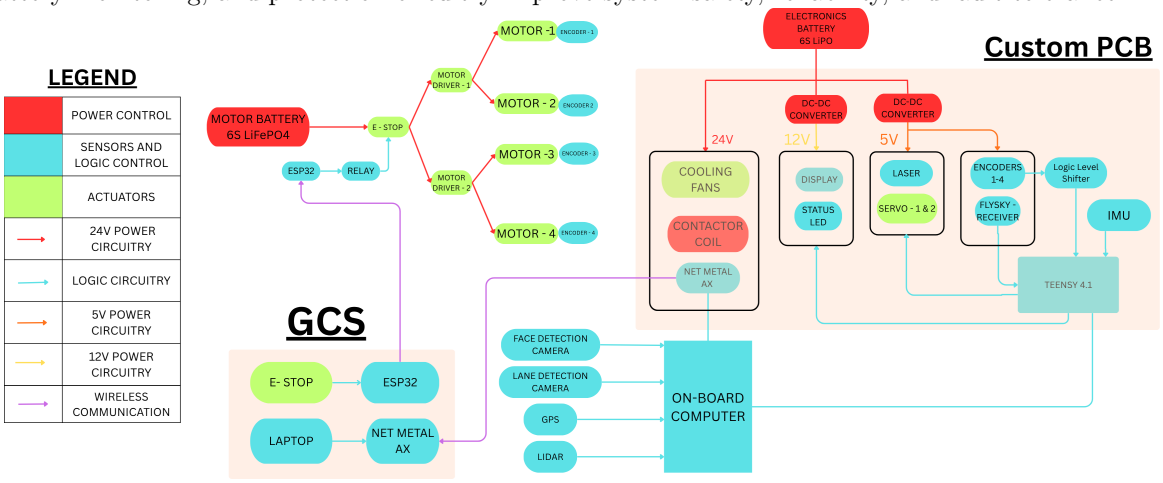


Figure 5: Electrical Architecture and Power Distribution Overview

4.2 Main Components

Component	Function / Description
Teensy 4.1 Microcontroller	**Real-time embedded controller** responsible for **motor control**, encoder processing, **PID velocity control** and hardware interfacing.
Onboard Computer	Executes **ROS 2**, autonomous navigation, perception, localization, mission planning, and communicates with the Teensy 4.1 via **serial interface**.
Integrated Power & Logic Control Board	Integrates **power regulation**, electrical distribution, signal conditioning, embedded processing, and **system protection**.
NetMetal ax Wireless Access Point	Provides **robust wireless communication** and network connectivity.
RPLIDAR A3M1	Provides continuous **360° laser scanning** for mapping, localization, and obstacle detection.
Adafruit BNO085 IMU	Supplies fused **orientation**, angular velocity, and acceleration measurements for **motion estimation**.
4 × 210 RPM Planetary Geared DC Motors	Provides **differential-drive propulsion** with integrated **4784 CPR quadrature encoders** for closed-loop velocity control.
2 × Cytron MDDS30 Motor Drivers	Drives the left and right motor pairs with **regenerative braking**, **current limiting**, and thermal protection.
2 × MG995 Servos	Provide **270° pan** and **180° tilt** motion for camera and laser positioning.
ESP32 Wroom 32U	Implements the **wireless emergency stop** functionality.
Albright SD150A Contactor	Provides **electrical isolation** and safe switching of drivetrain power.
Status LEDs & Laser Module	Provide **system status indication** and **laser aiming** functionality.

4.3 PCB and Power Distribution

The custom **Power Distribution and Logic Control Board (PDU/PCB)** serves as the electrical backbone of the rover by integrating **power management, embedded control, signal conditioning, and subsystem interfacing** into a single compact platform. Its mixed-signal design, optimized routing, dedicated ground planes, and high-current traces minimize **electromagnetic interference (EMI)** while improving reliability and reducing wiring complexity.

The PCB distributes power through **regulated and fused outputs**, maintains electrical isolation between the electronics and drivetrain batteries, and houses the **Teensy 4.1** microcontroller for real-time motor control, encoder processing, PID computation, and peripheral communication. Integrated **voltage/current monitoring**, logic-level translation, industrial-grade connectors, and protection circuitry enhance fault diagnosis, operational safety, and long-term reliability, providing a scalable platform for autonomous rover operation.

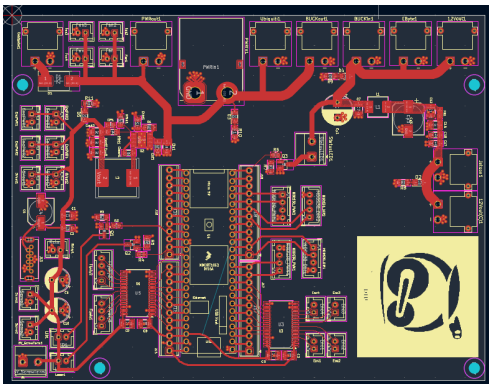


Figure 6: PCB

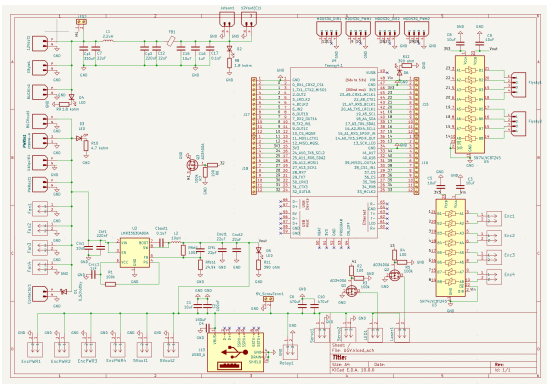


Figure 7: SCHEMATIC DIAGRAM

4.4 Manual Drive System

The rover includes a dedicated **manual driving system** for testing, teleoperation, and emergency recovery. Control is provided by the **FlySky FS-i10B** 2.4 GHz transmitter and receiver, which deliver reliable, low-latency communication between the operator and the onboard controller. The controller processes throttle, steering, and auxiliary inputs to generate motor commands. The FlySky FS-i10B uses the **AFHDS (Automatic Frequency Hopping Digital System)** to ensure stable, interference-resistant communication in outdoor environments. As a redundant control system, it allows operators to immediately take control during autonomous software failures, communication loss, or sensor faults, improving operational safety and mission reliability.

4.5 Emergency Stop Systems

Two independent, hardware-level emergency stop (E-stop) mechanisms are implemented to guarantee immediate vehicle immobilization under all fault conditions. To ensure maximum reliability and completely bypass potential software lockups within the primary autonomous navigation stack, both E-stop pathways converge directly at the heavy-duty Albright SD150 main power contactor:

- **Mechanical E-stop:** A high-visibility, normally-closed (NC) manual disconnect switch is integrated directly into the main traction contactor. Depressing this actuator physically forces the main contacts open, instantly severing the high-current path to all motor drivers. The switch is ergonomically positioned at the vehicle's centre-rear, complying strictly with the 60–120 cm height requirement to guarantee rapid track-side access.
- **Wireless E-stop:** A dedicated, highly robust RF link provides remote immobilization capabilities ($\geq$200 m range), entirely decoupled from the vehicle's primary ROS 2 network. This system utilizes a pair of ESP32 microcontrollers communicating via the ESP-NOW protocol configured in Long Range mode, ensuring low-latency, MAC-to-MAC connectivity without the overhead or vulnerability of a standard Wi-Fi network. Upon triggering the remote switch, the receiving ESP32 commands an isolated relay module to immediately cut power to the SD150's auxiliary coil. De-energizing this coil forces the contactor open, achieving a definitive hardware-level vehicle halt that cannot be defeated by an autonomous software failure or main microcontroller crash.

4.6 Communication System

The rover employs a **long-range wireless communication system** using the **MikroTik NetMetal AX** radio and **Ubiquiti airMAX AM-2G15-120** sector antenna to provide reliable connectivity between the **Ground Control Station (GCS)** and the rover for telemetry, **ROS** communication, HD video streaming, and remote control during outdoor missions.

The **NetMetal AX** offers industrial-grade, weather-resistant operation, while the **AM-2G15-120** is a dual-polarized **2×2 MIMO** antenna operating in the **2.4 GHz** band with **15 dBi** gain and approximately **120°** sector coverage, ensuring robust long-range communication.

5 Software Design

5.1 Overview

The software stack is built on ROS 2 Jazzy running on Ubuntu 24.04 and follows a *simulate extensively, deploy physically* methodology, with every subsystem validated in Gazebo Harmonic before deployment on physical hardware. The architecture spans seven functional layers,Infrastructure, Sensing, State Estimation, Perception, Planning, Actuation, and Control,implemented using a modular ROS 2 package structure. Hardware-dependent interfaces (motor control, turret communication, and GPIO peripherals) are isolated in dedicated packages, allowing the same localisation, planning, and decision-making software to operate identically in both simulation and physical deployment. Development is managed through a branch-per-subsystem workflow along with Docker-based reproducible environments.

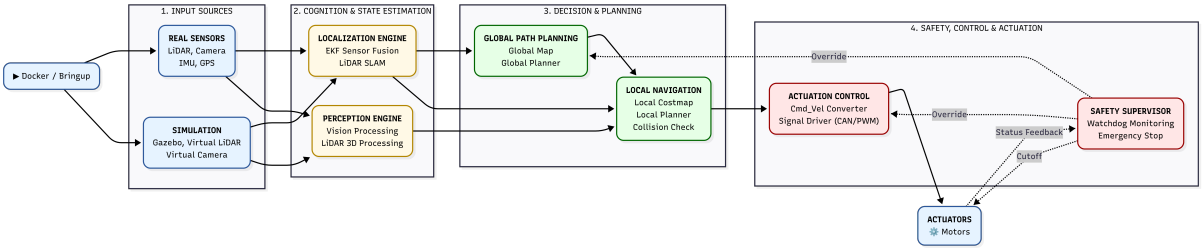


Figure 8: General Software Architecture

5.2 Core Functionalities

5.2.1 Mapping, Lane Detection, and Obstacle Avoidance

Localisation is provided by SLAM operating in asynchronous mode, constructing a 2D LiDAR occupancy map and publishing the `map`→ `odom` transform. The SLAM-generated map is intentionally excluded from the Nav2 global costmap, as its finer resolution caused oscillatory replanning when combined with perception costmaps during validation. Robot pose estimation is refined by an EKF operating at 50 Hz in planar mode, fusing wheel odometry, IMU measurements, and GPS-derived odometry for locally smooth, drift-tolerant state estimation. The global planner operates on a four-layer semantic costmap: Lane boundaries are projected onto the ground plane and inserted directly into the Nav2 costmap as lethal regions, so the planner naturally generates lane-compliant trajectories without any specialised planning algorithm. Pothole detection accepts contours based on shape compactness and aspect ratio thresholds, then projects them onto the ground plane via ray casting through the TF tree. An ID-based deduplication mechanism assigns each pothole a permanent identifier on first detection; subsequent observations of the same physical pothole are ignored, preventing costmap growth as the vehicle approaches. Detected potholes are published as persistent lethal regions with no temporal decay.

5.2.2 Waypoint Navigation

Three competition waypoints (WP-1, WP-2, WP-3) are preloaded from a YAML file in world coordinates. Mission execution is governed by a finite-state machine, with successful arrival at each waypoint triggering deterministic state transitions. A waypoint monitoring component tracks vehicle progress at 10 Hz and declares successful arrival when the rover enters a 0.5 m acceptance radius, publishing a structured event to `/waypoint_reached`. Reaching WP-1 advances the mission to the next navigation phase, WP-2

initiates the face-recognition task, and completion of the recognition sequence (or expiry of its time budget) transitions the system back to navigation towards WP-3. To improve navigation reliability, the perception pipeline continuously extracts the forward drivable region from the bird's-eye-view lane representation and generates a local navigation goal ("carrot") several metres ahead of the vehicle. Candidate goals are evaluated against lane boundaries, occupancy costmaps, and obstacle constraints before the safest forward goal is selected. If no suitable goal is available, progressively more conservative fallback strategies generate feasible forward targets, ensuring uninterrupted navigation. This receding-horizon process repeats continuously as the vehicle advances, allowing the rover to follow lane geometry while adapting to changing obstacles and road curvature. Global path planning uses the layered semantic costmap via a sampling-based planner; local trajectory execution is handled by a critic-based controller with weights prioritising path fidelity over goal proximity.

5.2.3 Image Detection and Recognition

Face recognition uses an embedding-based model combining an ArcFace metric learning approach with a deep residual backbone. A normalised reference embedding is generated at startup from the target image using the largest detected face. At each of the 21 scan positions of the 7×3 boustrophedon search grid, the recognition component processes a captured image, computes embeddings for all detected faces, and evaluates cosine similarity against the reference. A detection is declared when the similarity score exceeds a configured threshold, validated against all six simulation face images. Confirmed detections trigger a proportional visual-servoing controller that refines turret alignment before activating the laser. A 45 s timer limits the total face-search duration and automatically advances to the next waypoint when the time budget is exhausted, guaranteeing forward mission progress regardless of detection outcome.

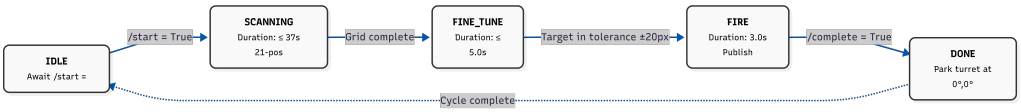


Figure 9: Face Detection Pipeline

5.3 Auxiliary Systems

Motor Control. Wheel actuation uses the `ros2_control` framework with a differential drive controller accepting stamped velocity commands. A lightweight bridging component provides a consistent interface between the navigation stack and low-level control, allowing identical motion-control logic in both simulation and hardware. The turret controller automatically switches between the Gazebo interface (simulation) and serial commands to the onboard microcontroller (hardware). GPIO-controlled LEDs display solid during initialisation, flash during autonomous operation, and return to solid on mission completion.
Vehicle Monitoring. A dedicated monitoring stack supervises system health throughout the mission. The health monitor checks topic liveness for key sensing and control interfaces at regular intervals using a configurable silence threshold, and verifies node availability against a predefined set of mission-critical nodes. Essential TF relationships (`odom` → `base_link`, `map` → `odom`) are validated continuously; structured JSON diagnostics are published to `/system_status` and `/system_alerts`. A dashboard publisher generates human-readable summaries at 1 Hz, and the telemetry logger records system, alert, state, control, navigation, resource, encoder, and power data to parallel JSON logs for post-mission analysis.
Sensor Fusion. The EKF operates at 50 Hz in planar mode, fusing wheel odometry (relative position, velocity, and yaw), IMU (orientation and angular rates with gravity compensation), and GPS (absolute x-y position). The resulting `odom` → `base_link` transform is consumed by Nav2 for locally smooth state estimation while allowing global corrections through the `map` frame.

5.4 System Integration and Fault Tolerance

The software stack is deployed through separate simulation and hardware launch configurations built on a shared base launch, with mission execution managed independently. Only the control and hardware interface packages differ between deployments, allowing the perception, localisation, and planning stacks to remain unchanged. Navigation failures invoke a reverse-and-reorient recovery behaviour, while a mission watchdog advances execution to WP-3 if the face-search task exceeds its time budget.

Watchdog Monitor. The *watchdog_monitor* package provides runtime supervision through three ROS 2 executables: *health*, *waypoints*, and an optional terminal *dashboard*. The health node validates the ROS computation graph against a predefined node set while monitoring odometry, LiDAR, and IMU topic liveness using timeout-based checks, publishing consolidated system health and diagnostic alerts. The waypoints node performs TF2-based mission tracking by querying the `map`→`base_link` transform to detect waypoint arrival and publish mission progress, supporting automatic waypoint re-arming for repeated traversals. The dashboard subscribes to these diagnostics and presents a lightweight real-time view of node status, alerts, and waypoint progression for operator monitoring.

Recovery Handling. The recovery module is decoupled from the lane pipeline via a *recovery-active* flag. Stuck detection fires only when low map-frame displacement is confirmed alongside an obstacle ahead (cross-checked against LiDAR and the local costmap), avoiding false triggers on a stationary robot in clear space. Rather than relying on default Nav2 BackUp/Spin behaviours, which abort on contact, the rover drives controlled reverse increments via direct velocity commands, gating each step on local costmap clearance. Once clear, it scans candidate headings scored on goal alignment, lateral clearance, and free-path distance, committing to the best heading above a minimum quality threshold. If no suitable heading is found, the back-up distance is escalated up to a bounded number of attempts before the robot holds position facing the goal. After committing, the module waits for several consecutive confirmations of a clear path before returning control to the lane-following pipeline. Any new navigation goal immediately cancels recovery and resets state.

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5.5 Software Testing and Bug Tracking

Each subsystem in software development was independently verified in Gazebo Harmonic using the UGVC track environment prior to integration. Key validation scenarios included:

- Lane detection performance under varying illumination conditions
- Pothole deduplication across repeated traversals
- EKF drift assessment over full-course missions
- Face recognition across all six simulation face images
- End-to-end mission completion within the 8 min competition limit

Bug tracking was managed through GitHub Issues. Major issues resolved included pothole false positives from perspective distortion (mitigated by a shape-based filter), oscillatory replanning from mismatched map resolutions (resolved by excluding the SLAM map from the Nav2 costmap), and unstable long-range navigation (addressed through continuous BEV-based local goal generation and receding-horizon planning). Physical hardware testing subsequently validated all subsystems against simulation results. A dedicated turret dry-run mode enabled hardware-independent verification of targeting logic before integration with the physical pan-tilt platform.

6 Simulation and To-Date Testing

6.1 Simulation Environment

All development and validation testing was conducted in Gazebo Harmonic. The simulation world models the real competition course: a rectangular asphalt track approximately 160 m long with lane widths of 3–6 m, minimum turning radius 1.5 m, and white-line borders approximately 8 cm wide. Six face image models are placed around the Waypoint 2 zone, each displaying a distinct person photograph. The robot URDF is fully physics-simulated, including wheel friction, inertia tensors, mass distribution, and sensor noise models (Gaussian noise on GPS, quantisation on LiDAR range).

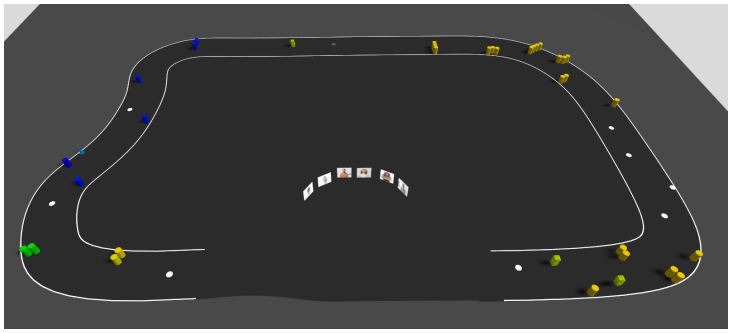


Figure 10: Simulated UGVC Environment in RViz

6.2 Key Testing Scenarios

Scenario	Subsystem(s) Exercised	Observed Outcome
Lane-following on straight (15 m)	Lane costmap + BEV controller	Lane centre error $\leq \pm 5$ cm over 15 m
Pothole avoidance (60 cm circle)	Pothole costmap + Nav2 replanning	Robot routes around potholes; no contact
Waypoint navigation (all 3 WPs)	Mission manager + Nav2	All 3 waypoints reached with <1.0 m error
Face detection (all 6 sim photos)	Turret vision + ArcFace	Correct face identified in <15 s average
TF startup delay recovery	TF timeout handling	No silent frame drops; WARN logs generated
Start-occupied planner resilience	Reduced inflation + tolerance	No planner aborts near pothole inflation
Lane-costmap pothole exclusion	Shape compactness dual filter	No circular blobs leaked into lane costmap

6.3 Visual Results

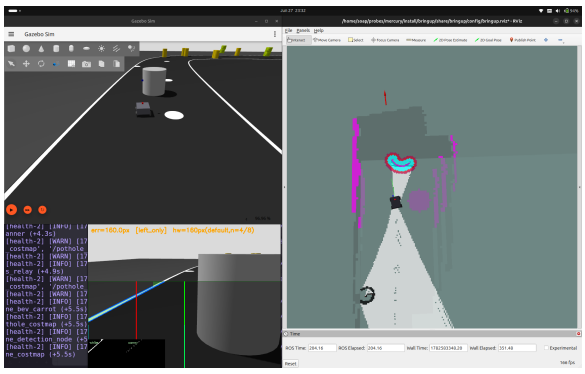


Figure 11: Lane detection simulated in RViz

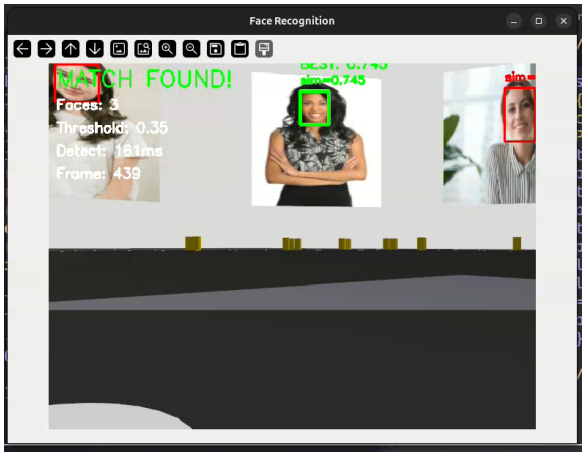


Figure 12: Working of face detection mechanism

6.4 Performance Metrics

Metric	Simulation Result	Competition Requirement / Target
Waypoint arrival error (all 3 WPs)	<1.0 m	$\leq$1.5 m

Metric	Simulation Result	Competition Requirement / Target
Face detection time (avg, 6 photos)	$<$15 s average	$\leq$60 s (hard limit)
Lane-keeping error – straight	$\pm$5 cm from centre	No border crossing
Pothole detection latency	$<$200 ms after FoV entry	Before robot reaches obstacle
Navigation speed (simulation)	0.6 m/s (2.16 km/hr)	2–10 km/hr competition requirement
EKF output rate	50 Hz	$\geq$10 Hz for Nav2
System startup	$<$8 s (simulation)	$<$15 s (real hardware target)
Face task worst-case time	21×2.2 s = 46.2 s	$<$60 s

6.5 Real-Time Testing Environment

Real-world hardware validation is currently underway and will continue until the team's departure for the competition. While the complete autonomy stack has been extensively validated in **Gazebo Harmonic**, ongoing field trials are focused on verifying end-to-end autonomous operation on the physical rover, correlating simulation results with real-world performance, and refining the system for reliable competition deployment.

To validate robustness, the complete autonomy pipeline was evaluated on **four different Gazebo maps** with varying track layouts, lane widths, turn radii, and pothole placements. The rover successfully completed each course autonomously with consistent lane-following and waypoint navigation performance, demonstrating good generalization beyond a single competition map.

Current hardware validation activities include:

- **Homography calibration** for accurate BEV generation.
- **Lane detection tuning** under outdoor lighting.
- **Speed calibration** to achieve reliable operation within the competition speed range of **2–10 km/h**.
- **GPS/EKF integration** for improved outdoor localization and reduced drift.

6.6 Challenges Encountered and Risk Management

- **Reactive costmap horizon:** At startup, the semantic costmap is limited to the currently perceived lane region. Continuous BEV-based local goal generation enables the rover to navigate using locally valid goals while the perception pipeline progressively expands the drivable area ahead.
- **Pothole silent misses:** A morphological close kernel was inadvertently commented out during refactoring, causing fragmented blobs to fall below the area threshold. Corrected by reinserting the kernel and adding a unit test to the perception package.
- **Noon outdoor lighting:** Colour-space thresholds are exposed as ROS 2 parameters adjustable without recompilation. A calibration session in noon lighting is planned before the competition.
- **Qualification speed requirement:** The competition mandates $\geq$2 km/hr. Raising the maximum velocity to 0.6 m/s is flagged as a critical pre-competition action item.
- **SLAM drift over 160 m:** Long-course SLAM accumulates yaw drift. GPS/EKF integration provides absolute position correction; SLAM loop-closure activates on return near the start line.

7 Analysis of Complete Vehicle

7.1 Unique Software, Sensor, and Control Strategies

Avni's autonomous system is specifically designed to tackle the unique demands of this year's MTC Cairo ground-vehicle challenge. Its sensor fusion framework integrates a GenX RTK F9P Tough Ground RTK GPS for absolute localisation, RPLiDAR A3M1 for 360° obstacle detection, a front-facing camera for lane perception and pothole detection, and a turret camera for face recognition. A key innovation is the reactive lane costmap strategy, where lane boundaries detected by the vision system are injected into the Nav2 costmap as lethal cells. This allows the standard global and local planners to naturally guide the robot within lane boundaries without requiring a separate lane-following algorithm, significantly simplifying control logic and improving adaptability.

On the software side, Avni features a modular ROS 2-based architecture combining Nav2 planning with a BEV-based receding-horizon local goal generation layer. The global planner generates lane-constrained paths, while the local planner continuously tracks dynamically generated forward navigation

goals derived from the bird's-eye-view lane representation. The mission management component coordinates waypoint transitions, with a 55 s watchdog ensuring the mission always advances even under face-search failure. All modules were rigorously validated in a Gazebo Harmonic virtual environment before field deployment.

7.2 Initial Performance Assessments

Avni's simulation-validated performance against the competition's key requirements is summarised below. In the initial testing phase, each component and subsystem was individually verified in simulation and passed all predefined performance criteria.

1. The EKF-fused localisation stack was validated at 50 Hz, well above the Nav2 minimum of 10 Hz.
2. A maximum forward velocity of 0.6 m/s (2.16 km/h) was validated in simulation, satisfying the competition's minimum speed requirement while maintaining stable autonomous navigation.
3. The complete 21-position face search completes in a worst-case of 46.2 s, within the 60 s hard limit.
4. Waypoint arrival error was below 1.0 m across all three waypoints in simulation.
5. The lane-keeping error on straight segments was ± 5 cm from the lane centre.
6. Pothole avoidance was validated with no contact events across all test runs.
7. System startup (all components live) was achieved in under 8 s in simulation, with a 15 s target for real hardware.

8 Future Work

The following improvements are planned and will be completed before the competition :

- **GPS/EKF outdoor integration:** Enable the GPS sensor in the EKF configuration; fuse GNSS and SBAS absolute position with SLAM localisation to eliminate long-term yaw drift over the 160 m course.
- **Real-hardware camera calibration:** Execute the homography calibration tool on the physical camera mount under outdoor noon lighting; update the homography matrix for the BEV controller; validate lane and pothole detection under Cairo conditions.
- **Pre-built static lane map:** Record a training-run costmap on the real competition track; replay it as an additional static layer so the global planner has complete lane boundary information from the first metre of the competition run.
- **LiDAR-camera obstacle fusion:** Project front-camera colour data onto LiDAR point clusters to detect non-white coloured obstacle barrels that the current pothole mask cannot identify.
- **Dynamic speed adaptation:** Implement a velocity scaling component that reduces speed near potholes and tight turns and restores maximum speed on clear straights to minimise net competition time.
- **Planning config cleanup:** Populate or remove the currently empty planning controller configuration to avoid judge queries during the code review stage.
- **Wireless E-stop range validation:** Test wireless E-stop at 150 m range before competition day; replace the module if range is insufficient.

9 References

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